

KAARA

AI USE CASE BUILD FACTORY

POC APPROACH DOCUMENT

AI-Powered Claims Processing and Settlement

Hypothesis ID: HYP-IN-001 | Industry: Insurance

Document	POC Approach & Productionisation Plan
Use Case	AI-Powered Claims Processing and Settlement
Hypothesis ID	HYP-IN-001
Industry / Sub-industry	BFSI / Insurance
Anchor Client (Synthetic)	TrustShield Insurance
Primary Business Impact	Ops Efficiency
Secondary Impacts	Bottom Line, Revenue Growth, Team Productivity
Delivery Platform	Kaara.Code + Compounding Build
Target POC Duration	10 weeks (Discovery → Demo → Pilot)
Target Production Rollout	14 weeks post-POC acceptance
Document Owner	Kaara Solutioning & Delivery

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1. Executive Summary

This document sets out the POC approach, productionisation plan, and commercials for AI-Powered Claims Processing and Settlement (HYP-IN-001), anchored on TrustShield Insurance — Mid-large Indian insurer with claims-AI and retention mandate. The POC targets measurable, executive-visible business impact inside a 10-week window, and lays the foundation for a full 14-week production rollout that follows.

TrustShield's claims TAT averages 12 days (motor), 18 days (health); leakage estimated at 8-11%. STP rate is 14%. 480-person claims team handles rising volume at flat productivity.

Our guiding hypothesis — Insurers with AI claims and enterprise memory will achieve 30–40% faster settlement within 18 months — is underpinned by four solution pillars (ingestion and memory, core domain model, governance, and business-outcome loop) that together exploit the fact that this is a compound-ready problem. In concrete terms, claims pattern intelligence compounds with every settlement: every data point generated during the POC enriches the enterprise memory layer and compounds into durable business advantage.

Commercially, the POC is priced at ₹58 L–72 L fixed + success-fee. The production commercial envelope is ₹3.0 Cr–4.2 Cr for Year 1. Expected outcome: ₹85-130 Cr annual — leakage + ops cost + retention uplift.

2. Business Context & Problem Statement

2.1 Why This Matters Now

TrustShield's claims TAT averages 12 days (motor), 18 days (health); leakage estimated at 8-11%. STP rate is 14%. 480-person claims team handles rising volume at flat productivity.

The business impact footprint is real and measurable. TrustShield Insurance operates at Combined general + life insurer; 12M policies in force; 68 branches, 2,400 agents; health + motor + life product lines — the scale at which small accuracy, cycle-time, or leakage improvements translate to significant bottom-line and top-line effects. At the same time, the insurance sector's macro dynamics (regulatory pressure, competitive digitisation, and the compounding effect of rival AI adoption) make inaction a negative-net-present-value choice over an 18-24 month horizon.

2.2 The Cost of Inaction

Inaction is not neutral. Across peer companies who have deferred AI investment on this problem, we observe: (a) steady degradation of the core operating metric over 12-24 months as competitors deploy AI, (b) rising headcount requirement just to keep pace with volume growth, (c) increasing regulatory and compliance drag, and (d) reduced ability to attract and retain top talent, who increasingly expect modern AI-augmented tooling.

2.3 What Good Looks Like

A well-instrumented AI-native capability in this area should produce four observable outcomes: (a) a measurable step-change in the primary business metric; (b) cycle-time compression from manual / sequential to near-real-time; (c) a human workforce moved up the value chain from data entry / routine to exception handling / strategy; and (d) an institutional knowledge base that makes the operation demonstrably better each quarter rather than merely proportional to volume.

3. Hypothesis and Target Outcomes

3.1 Guiding Hypothesis

Insurers with AI claims and enterprise memory will achieve 30–40% faster settlement within 18 months

3.2 POC-level Quantitative Targets

Metric	Baseline (TrustShield)	POC Target (W10)	Production Y1 Target
Motor-claims TAT	12 days	3 days	≤24 hrs
Health-claims TAT	18 days	7 days	≤3 days
STP rate	14%	42%	≥65%
Leakage reduction	baseline	–25%	–50%

Customer NPS on claims	baseline	+15 pts	+30 pts
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3.3 Qualitative Signals

- Executive sponsor can see a real-time dashboard that ties the metric back to business outcomes.
- Operational team spends more time on exception handling and strategy than on data re-entry.
- Every decision and outcome is captured as training signal for the next iteration.
- Risk, bias, and audit artefacts are retrievable for any decision without delay.

4. Anchor Client Profile — TrustShield Insurance

This POC is anchored on TrustShield Insurance (Mid-large Indian insurer with claims-AI and retention mandate). TrustShield is headquartered in Pune, Maharashtra with ₹4,500 Cr GWP (FY25-26) revenue, and operates as Combined general + life insurer; 12M policies in force; 68 branches, 2,400 agents; health + motor + life product lines.

4.1 Strategic Context at TrustShield

Key pain points that make this use case urgent: Claims TAT and leakage; fraud in motor & health; underwriting accuracy on new-age risks; retention of mid-tenure policies.

Strategic priorities directly aligned with this POC: Straight-through processing 70%+ on simple claims; 40-50% faster fraud detection; 20-25% retention lift; dynamic pricing in motor.

4.2 Systems & Data in Scope

Systems-of-record relevant to the POC include: Majesco Policy, ClaimCenter (Guidewire), custom fraud rules, SAP S/4HANA, Qlik BI, ServiceNow.

Pre-built synthetic datasets for realism and demo: claims.csv, policies.csv.

4.3 Decision Makers & Sponsorship

CEO: —.

5. Solution Approach

The Kaara solution is organised into four functional pillars, each of which becomes an independent workstream during delivery.

5.1 Pillar — Intake Intelligence

Multi-channel intake (app, WhatsApp, call) with instant FNOL capture, damage-photo AI, and document auto-read.

5.2 Pillar — Adjudication Automation

Rules + ML adjudicate simple claims end-to-end; complex claims routed to adjuster with pre-work done.

5.3 Pillar — Leakage Prevention

Pattern detection on over-billing, inflated repair quotes, and duplicate submissions.

5.4 Pillar — Settlement Memory

Every settled claim, every adjudication decision, every dispute outcome enriches the model.

5.5 Where Humans Stay in the Loop

This solution preserves a human-in-the-loop envelope throughout. High-value, high-risk, and low-confidence cases route to human review; routine, high-confidence cases flow through the automated path with full audit. This protects TrustShield's operational integrity and regulatory defensibility while releasing the majority of the productivity benefit.

5.6 Compounding Build Lens

Claims is a compounding problem — every settled claim teaches the system something. Kaara's Memory layer makes TrustShield's claims operation structurally faster, cheaper, and more accurate over time.

6. Reference Architecture

The reference architecture is organised into six layers. All layers deploy inside the customer VPC; data never leaves the country; all direct identifiers are tokenised at ingestion.

Layer	Components	Technology
L1 — Source Systems	Majesco Policy, ClaimCenter (Guidewire), custom fraud rules, SAP S/4HANA, Qlik BI, ServiceNow	Existing customer infrastructure
L2 — Ingestion & ETL	Event-driven + batch ingestion; normalisation to canonical domain model	Kafka (MSK), AWS Glue, Airbyte, domain-specific normalisers
L3 — Compounding Memory	Indexed semantic store, rule registry, lineage graph	pgvector on RDS, OpenSearch, Neo4j
L4 — AI Services	Domain-specific models implementing the 4 solution pillars	Kaara.Code orchestration; fine-tuned LLMs + classical ML
L5 — Workflow & HITL	User consoles, exception queues, SLAs	Kaara Workbench (React), Temporal workflow
L6 — Observability, Audit, Governance	Model cards, decision audit, vault, compliance	Kaara GovOps, CloudWatch, Okta, HashiCorp Vault

6.1 Deployment Targets (POC)

- Deployed inside TrustShield's AWS VPC (Mumbai region).
- Kubernetes (EKS) for stateless services; managed data layer.
- GitOps deployment with canary + instant rollback.
- Zero-trust service mesh; secrets via HashiCorp Vault.

7. Data Strategy & Source Systems

7.1 Data Required for POC

Data Source	Volume	Freshness	Sensitivity
Claims + policy DB	POC sample (24 months)	Daily or near-real-time	Tokenised + audited
Damage photos + invoices	POC sample (24 months)	Daily or near-real-time	Tokenised + audited
Medical reports + hospital bills	POC sample (24 months)	Daily or near-real-time	Tokenised + audited
Fraud-indicator history	POC sample (24 months)	Daily or near-real-time	Tokenised + audited
Adjuster notes	POC sample (24 months)	Daily or near-real-time	Tokenised + audited

7.2 Data Quality Remediation

A 3-week data remediation workstream runs in parallel with solutioning:

1. Field-level profiling across the data window to quantify missingness, outliers, and drift.
2. Schema harmonisation across source systems — resolving overlapping fields with conflicting semantics.
3. Tokenisation of sensitive identifiers in a one-way vault with reverse-lookup available only to authorised reviewers under audit.
4. Gold-standard corpus curation by domain experts to serve as the fine-tuning anchor.

7.3 Data Governance

Data remains in TrustShield's AWS tenant throughout. Kaara engineers access through bastioned, just-in-time VDI with queries logged to immutable audit stream. Data contracts are explicit on permitted uses and retention.

8. AI/ML Model Strategy

8.1 Model Portfolio

Each solution pillar is implemented by one or more models with explicit class, base-model path, and role. Model selection favours smaller, fine-tuned, explainable models for governance-critical decisions, and reserves large-LLM approaches for open-ended or generative tasks.

8.2 Evaluation Framework

Every model has a pre-agreed acceptance threshold before entering shadow mode, and a second threshold before it takes production load. A blinded panel evaluates a weekly sample of decisions. Pass-bar for shadow is $\geq 90\%$ domain-expert agreement; for production it is $\geq 96\%$ on decisions within the autonomy band, with HITL safety net on anything below.

8.3 Bias, Fairness, and Safety

Particular attention is paid to segment-specific bias vectors relevant to the use case. Each vector is tested quarterly against a curated bias-probe set, and disparity above a pre-agreed threshold triggers a model review.

9. Integration Strategy

The inbound and outbound integration plan is prioritised by both value and effort.

System	Direction	Mechanism	POC Notes
Guidewire ClaimCenter	Bidirectional or directional as relevant	REST / HL7 / file-drop / EDI as applicable	POC sandbox first
Majesco Policy	Bidirectional or directional as relevant	REST / HL7 / file-drop / EDI as applicable	POC sandbox first
Network-hospital APIs (health)	Bidirectional or directional as relevant	REST / HL7 / file-drop / EDI as applicable	POC sandbox first
Workshop-estimate platforms (motor)	Bidirectional or directional as	REST / HL7 / file-drop / EDI as applicable	POC sandbox first

	relevant		
UPI + NEFT for payouts	Bidirectional or directional as relevant	REST / HL7 / file-drop / EDI as applicable	POC sandbox first

Integration priority: (1) sources that feed the Compounding Memory layer with highest volume and signal; (2) action surfaces where the AI produces business outcomes; (3) audit and observability integrations for governance. A fallback path (RPA or manual) is retained for any integration in which the real-time API path degrades.

10. Governance, Compliance & Ethics

10.1 Regulatory Surface

- DPDP 2023
- RBI Master Directions applicable to the product
- SEBI regulations (where securities / investor-facing)
- IRDAI (where insurance-facing)
- FEMA where cross-border
- PMLA + KYC norms

10.2 Governance Operating Model

A three-tier governance stack operates alongside the solution: an Operating Council chaired by the customer sponsor, a domain-and-Compliance Board chaired by the domain executive, and a technical AI Safety Cell run by Kaara. Every model release is signed off by two of the three; every production deployment requires a model card, a bias report, and a rollback plan.

10.3 Ethics Principles

- AI proposes, human decides — no autonomous action outside a pre-agreed autonomy envelope.
- Full auditability — every decision has retrievable rationale, inputs, and model version.
- Explainability by default — plain-language rationale for every individual decision on request.

- Data stewardship — customer data is and remains TrustShield's; Kaara treats it with care and contractual obligation.

11. POC Scope, Plan & Milestones

11.1 In-Scope / Out-of-Scope

Dimension	In Scope (POC)	Out of Scope (Production / Phase-2)
Core focus	AI-Powered Claims Processing and Settlement on selected POC footprint	Broader adjacent use-cases (Phase-2)
Systems	Priority integrations from Section 9	Tail-system integrations and deep customisation
Users	Target personas defined in companion PRD	Full user population (production)
Scale	Bounded: representative slice of customer footprint	Full enterprise scale (production)

11.2 10-Week POC Timeline

Week	Workstream	Deliverable
W1-2	Mobilisation & Discovery	Data contracts, vault setup, VPC access, persona interviews, baseline metrics frozen
W3-4	Data ingestion & gold corpus	Historical data ingested, gold-standard corpus curated, embedding index live
W5-6	Model fine-tune & workbench build	Solution-pillar models at shadow-quality; user workbench usable
W7	Shadow mode	All pillars running in shadow against live data; zero live-side action

W8	Human-in-the-loop console	End-user console and exception queues; dashboards for sponsor
W9	Production-band cut-over	Pillars active for a bounded volume slice under success-gate
W10	Demo, read-out, decision gate	Executive demo; signed acceptance or go/no-go for production

11.3 POC Team

Role	Kaara FTE	TrustShield FTE
Engagement Partner / Architect	0.5	—
Domain SME (Senior)	1.0	1.0 (customer clinical/domain lead)
AI/ML lead	1.0	—
Data engineer	1.0	0.5
Integration engineer	1.0	0.5 (system SME)
Full-stack developer	1.0	—
Governance & compliance	0.3	0.3 (sponsor office)
Programme manager	0.5	0.5 (sponsor office)
Total (indicative)	~6.3	~2.8

12. POC Success Metrics & Acceptance Criteria

The POC is considered successful if, by end of Week 10, the quantitative targets in Section 3.2 are on track and the acceptance criteria in the companion PRD are jointly signed off. 'Must-have' criteria form the floor; 'stretch' criteria trigger an accelerated production rollout.

13. Demo Storyboard

The end-of-POC demo is anchored on a realistic end-to-end narrative with the sponsor, domain executive, and one external stakeholder in the room.

5. Opening frame — baseline vs POC dashboard side-by-side; key metric moving in real time.
6. Live vignette — a realistic scenario flows through the system end-to-end; AI proposes, human reviews, outcome recorded.
7. Deliberate-risk moment — a challenging / edge case demonstrates the guard-rails and human-in-loop design.
8. Governance view — every step so far is replayable in the audit console with rationale and model version.
9. Compound effect — a chart of model / system quality improving week-by-week as Memory accumulates.
10. Executive ask — signed acceptance of POC; kick-off of the 14-week production rollout.

14. Productionisation Plan

Moving from POC to production requires hardening across five dimensions: scale, reliability, security, operating model, and learning loops. The production plan is staged over fourteen weeks so that each dimension is proven before the next load is added.

14.1 14-Week Production Roadmap

Phase	Weeks	Focus	Exit Gate
Hardening	1–3	Performance, DR, secrets, access review, penetration test	Infra audit sign-off; DR drill successful
Scale wave 1	4–6	Expand from POC slice to ~50% of footprint	Core metric trending as planned

Scale wave 2	7–10	Extend to full footprint	All units / entities live
Stabilise	11–12	Noise reduction, feedback loops, playbook refinement	Production-quality steady state
Handover & optimise	13–14	Transition to run-state; model-refresh calendar signed; commercials settled	Formal handover; Year-1 run-state begins

14.2 Hardening Checklist

- Load and chaos testing — 3x peak-volume synthetic load, randomised component failures.
- Third-party penetration test and VAPT certificate — retained annually.
- BCP/DR plan — full failover drill in a staged environment before production.
- Secrets management — HashiCorp Vault with 90-day rotation on all credentials.
- Access review — quarterly least-privilege audit across Kaara and customer IDs.
- Model registry and release gate — every production promotion requires model card, bias report, rollback plan.

14.3 Change Management & Adoption

A dedicated change workstream runs in parallel: role-specific training, adoption champions, and measurable value-capture narrative shared weekly with the sponsor. The explicit intent is to move the existing workforce up the value chain rather than to displace it.

14.4 What 'Done' Looks Like at End of Year 1

- Full production footprint live across TrustShield's relevant entities.
- Core business metric at the Production-Y1 target in Section 3.2.
- Compounding Memory layer is populated with meaningful volume of training signal, increasing monthly.
- Sponsor publishes a real-time dashboard at the executive committee or board, quarterly.

15. Scale-out & Rollout Plan

Once the main rollout stabilises, the platform becomes the foundation for adjacent plays. In Year 2, natural adjacencies include further use cases listed in the AI_UseCases_Master_Tracker tied to the same Compounding Memory layer. A playbook template, infrastructure blueprint, and change-management kit are productised so that each subsequent adjacency launch is a <90-day onboarding rather than a fresh build.

16. Operating Model, SLAs & Run-State

16.1 Run-State Team

Function	Role	FTE
Platform SRE	24x7 rota, alerting, incident mgmt	4.0 (distributed)
Model Operations	Drift monitoring, weekly retraining	1.5
Domain Operations	Shadow audit, exception triage, feedback loop	2.0 + embedded
Account & governance	Monthly governance council, quarterly review	0.5
Compliance & audit	Regulatory obligations, evidence pack	0.5

16.2 SLAs

- Core business-latency SLA (solution-specific; see companion PRD Part B).
- Platform uptime: 99.5% business hours; 99.0% overall.
- P1 incident response: 15 min acknowledgement, 2 hour resolution target, RCA within 72 hours.
- Model drift review: weekly signal, monthly formal; retraining within 10 business days of threshold breach.
- Security incident: notify DPO within 4 hours; notify affected parties per DPDP norms.

17. Commercials — POC and Production

Kaara's commercial posture is outcome-aligned: a fixed-price POC with success-fee on achievement of must-have criteria, followed by a production contract that blends platform licence, throughput fee, and outcome-linked annual rebate.

17.1 POC Commercial Envelope

Line Item	Amount	Notes
Fixed POC fee	₹58 L–72 L	Covers ~6.3 Kaara FTE for 10 weeks + tooling + infra
Infra reimbursement	At cost	Production-grade sandbox; billed on actual consumption
Success fee	Included above	Paid on joint acceptance of must-have criteria
Payment schedule	30/40/30	Kick-off / W6 / on acceptance

17.2 Production Commercials (Year 1)

Line Item	Amount	Structure
Platform licence	₹3.0 Cr–4.2 Cr	Annual, capped at defined footprint
Integration & rollout	Included in envelope	One-time, amortised over 14 weeks
Managed run-state	Included (Y1)	Transitions to co-managed from Y2
Business-outcome rebate	Up to 15% credit	Credit back if outcome targets exceeded

17.3 Expected Business Outcome

₹85-130 Cr annual — leakage + ops cost + retention uplift

18. Risks, Dependencies & Mitigations

Risk	Likelihood	Impact	Mitigation
Image-based damage assessment accuracy	Med	High	HITL on low-confidence; continuous-training with adjuster-labelled corrections
IRDAI regulatory scrutiny on STP	Med	High	Explainable decisions; TAT compliance; complaint-ratio monitoring
Provider (hospital/workshop) pushback	Med	Med	Transparent rationale; dispute-resolution SLAs; continued provider-engagement

19. Kaara Differentiators

- Compounding Build architecture — Enterprise Memory layer retains and compounds institutional knowledge; point-solution vendors reset with every staff or vendor change.
- Kaara.Code agentic orchestration — end-to-end agentic workflows rather than isolated APIs.
- Governance-first posture — DPDP-native design, model cards, audit trail, bias probes; makes Kaara a low-risk AI partner.
- Domain depth — in-house domain experts and advisor network.
- Outcome-aligned commercials — credible signal that Kaara is invested in business outcomes, not invoice volume.
- India-first, global-scale — deep understanding of the Indian regulatory and operational context.

20. Appendices

20.1 Linked Documents

- Product Requirements Document (PRD) — companion document for this use case.
- TrustShield Insurance Synthetic Data Specification — claims.csv, policies.csv.
- Kaara Compounding Build Architecture Brief.
- AI_UseCases_Master_Tracker_Kaara.xlsx — Hypothesis HYP-IN-001.